

Beyond Benchmark Performance in Ophthalmic AI: A Review and Evidence Gap Map

Tobias Schlosser^{1a}, Michael Friedrich^{1b}, Richard Vogel^{1c},
Benny Platte^{1d}, Matthias Vodel^{1c}, Danny Kowerko^{1b,d}, Marc Ritter^{1d},
Miriam Goldammer^{1a}, Vinodh Kakkassery^{1e}, Ramin Khoramnia^{1f}, Markus Wolfien^{1a,g}

^a Dresden University of Technology, Faculty of Medicine Carl Gustav Carus, Center for Medical Informatics, Fetscherstraße 74, Dresden, 01307, Saxony, Germany

^b Chemnitz University of Technology, Faculty of Computer Science, Junior Professorship of Media Computing, Straße der Nationen 62, Chemnitz, 09107, Saxony, Germany

^c University of Applied Sciences Mittweida, Faculty of Applied Computer and Biosciences, Professorship of Software Development for Media and Application Systems, Technikumplatz 17, Mittweida, 09648, Saxony, Germany

^d University of Applied Sciences Mittweida, Faculty of Applied Computer and Biosciences, Professorship of Media Informatics, Technikumplatz 17, Mittweida, 09648, Saxony, Germany

^e Klinikum Chemnitz, Department for Ophthalmology, Flemmingstraße 2, Chemnitz, 09116, Saxony, Germany

^f University Hospital Carl Gustav Carus Dresden, Department of Ophthalmology, Fetscherstraße 74, Dresden, 01307, Saxony, Germany

^g Dresden University of Technology, Center for Scalable Data Analytics and Artificial Intelligence (ScaDS.AI) Dresden/Leipzig, Strehlener Straße 12, 14, Dresden, 01069, Saxony, Germany

Summary

Ophthalmology combines detailed imaging with longitudinal measurements that inform screening, monitoring, and treatment, providing a distinctive setting for clinical artificial intelligence (AI). We reviewed 427 English-language publications involving optical coherence tomography (OCT), OCT angiography (OCTA), or related ophthalmic imaging, identified from 6,482 harmonised records retrieved from PubMed, ACM Digital Library, IEEE Xplore, and Scopus through April 2026. Applications extended beyond image classification to multimodal interpretation, prediction of disease course and treatment need, and clinical workflow support. Evidence mapping showed extensive attention to model performance but substantially less to implementation and its consequences, with further gaps in economic evaluation, regulation, cybersecurity, privacy, and accountability. Evidence in these domains provides a foundation for the next stage of clinical translation. Progress can build on this foundation through representative external validation, interoperable data, effective human oversight, and prospective evaluation of safety, equity, and patient and service outcomes. Evaluating ophthalmic AI across the care pathway is therefore essential to establish whether model outputs improve clinical decisions and deliver measurable benefits for patients and health services.

Keywords: Artificial Intelligence, Ophthalmology, Clinical Translation, Optical Coherence Tomography, Evidence Gap Map

1. Introduction

Vision loss can diminish independence, participation, and wellbeing, with time-trade-off evidence illustrating the value people place on preserving sight.¹ The eye's accessibility and wealth of imaging data make ophthalmology particularly well suited to digital research. Helmholtz's ophthalmoscope enabled examination of the living fundus in 1850–51, while subsequent advances in fundus photography, OCT, and OCTA enabled non-invasive measurement of neural and vascular structures.^{2–4} This imaging capability is particularly relevant to glaucoma, age-related macular degeneration (AMD), and diabetic retinopathy (DR), which are among the leading global causes of blindness in people aged 50 years or older. Diabetic macular oedema is a sight-threatening manifestation of DR, while retinal vein occlusion (RVO) is an important retinal vascular cause of vision loss.^{5,6} Across these conditions, OCT supports diagnosis, segmentation, biomarker quantification, prognosis, treatment-response prediction, and longitudinal monitoring.^{7–10} These cap-

abilities connect image-derived measurements with decisions throughout the course of care. Together, substantial disease burden, rich imaging data, and longitudinal decision-making make ophthalmology a compelling but demanding setting for AI.

Digital health technologies could extend ophthalmic AI from image interpretation to decision support and care-pathway optimisation. This includes teleophthalmology, remote monitoring, automated triage, image-quality assessment, multimodal imaging, patient-reported outcomes, electronic health record data (EHR), and large language models (LLMs).^{11–14} They could support earlier referral, more consistent interpretation, individualised monitoring, more efficient use of specialist capacity, and clearer patient communication, particularly for ageing populations as well as underserved communities facing eye-care workforce shortages. As these benefits remain largely prospective, their effects on access, workload, equity, safety, and cost require evaluation in clinical practice.

Benchmark performance describes how well a model performs a defined task, whereas clinical utility depends on whether

its use improves care. Performance on curated datasets may not generalise across devices, institutions, populations, disease spectra, image quality, annotation practices, and workflows. A systematic review and meta-analysis found lower performance in external than internal validation, with greater training-data diversity associated with better external performance.¹⁵ Clinical translation also depends on overcoming fragmented data, limited semantic interoperability, unrepresentative training data, insufficient transparency, and unresolved ethical, legal, and regulatory questions.^{11,16,17} Yet clinical value does not require matching subspecialists on every benchmark. AI may improve care by increasing consistency, identifying uncertain or high-risk cases, supporting generalists, and prompting timely specialist referral. Evaluation should therefore compare AI-supported care with the relevant existing pathway and measure outcomes that matter to patients and health services.^{18–20}

We therefore examined evidence and gaps across the clinical, technical, data-related, implementation, ethical, and governance dimensions of ophthalmic AI involving OCT, OCTA, or related imaging. Figure 1 provides the framework informing our roadmap for clinical translation. The accompanying **Ophthalmic AI Evidence Portal** provides interactive online access to publication-level evidence, enabling descriptive subgroup analyses by clinical context, imaging modality, AI method, task, and validation characteristics. Readers can combine filters and trace mapped findings to individual publications to explore evidence gaps relevant to specific clinical or methodological questions.

Roadmap for clinically valuable and accountable ophthalmic AI

1. **Data and protocol foundations.** Document data provenance, device context, longitudinal structure, ophthalmic concept definitions, and reference-standard procedures.
2. **Clinical validation and utility.** Specify the intended use, target population, care setting, and decision threshold, then evaluate effects on decisions, outcomes, workload, access, safety, equity, and costs.
3. **Workflow integration.** Design outputs around clinical actions such as referral, monitoring, follow-up intervals, treatment timing, or counselling, and define clinician oversight.
4. **Fairness and governance.** Evaluate subgroup and intersectional performance, quantify uncertainty, and address privacy, cybersecurity, auditability, regulatory requirements, and drift monitoring.
5. **Next-generation decision support.** Develop multimodal, language-model, or agentic functions only when intended use, human responsibility, uncertainty, escalation pathways, and post-deployment surveillance are explicit.

2. Methods

2.1. Search strategy and selection criteria

We searched PubMed, ACM Digital Library, IEEE Xplore, and Scopus for records available by 30 April 2026. Search concepts combined artificial intelligence, machine learning, and deep learning with classification, detection, segmentation, monitoring, prediction, and screening. Imaging and clinical terms included optical coherence tomography, OCT, OCTA, ophthalm*, retina*, macula*, glaucoma, diabetic retinopathy, and age-related macular degeneration. We included English-language reports that described, evaluated, or synthesised ophthalmic AI involving OCT, OCTA, or related imaging and addressed a clinically or translationally relevant task. Relevant tasks encompassed diagnosis or detection, classification, segmentation or quantification, monitoring, prognosis or treatment-response prediction, screening or triage, multimodal interpretation, and workflow support. The imaging criterion defined the review's scope, while eligibility across primary study designs allowed comprehensive mapping of the available evidence.

Three reviewers (T.S., M.F., and R.V.) applied the eligibility criteria, with at least two authors involved in each decision and disagreements or borderline cases resolved by consensus. Contextual sources used for methodological or interpretive purposes were excluded from evidence-map counts. The supplementary material provides publication-level mappings, a public-dataset overview, and an extended synthesis.

2.2. Evidence synthesis and mapping approach

Reporting was informed by the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 statement, with Campbell guidance informing evidence-map design and synthesis.^{21,22} Alongside descriptive and qualitative synthesis, we conducted keyword co-occurrence analysis, international co-authorship mapping, public-dataset mapping, and evidence gap mapping. We extracted publication type, ophthalmic condition or clinical context, AI methods and intended role, data-related challenges, implementation barriers, and ethical, governance, and regulatory issues. Evidence mapping linked each publication to one or more dimensions, with a single stance recorded for each dimension.

For each general dimension, assignments were positive when a publication presented AI as enabling or improving the dimension, negative when an AI-related barrier, risk, failure, or constraint was identified, and balanced when evidence supported both directions. In the ethics gap map, *caused* indicated that AI introduced or amplified an issue, *resolved* that an AI-related approach mitigated the issue, and *caused and resolved* that evidence supported both directions.

Assignments required an explicit statement or unambiguous result, method, limitation, discussion point, or conclusion linking AI to the relevant dimension and stance. Passing mentions, generic future possibilities, standard ethics approval or consent, data-availability statements, and uncontextualised hardware specifications were insufficient. Absent or ambiguous evidence received no assignment rather than a neutral classification.

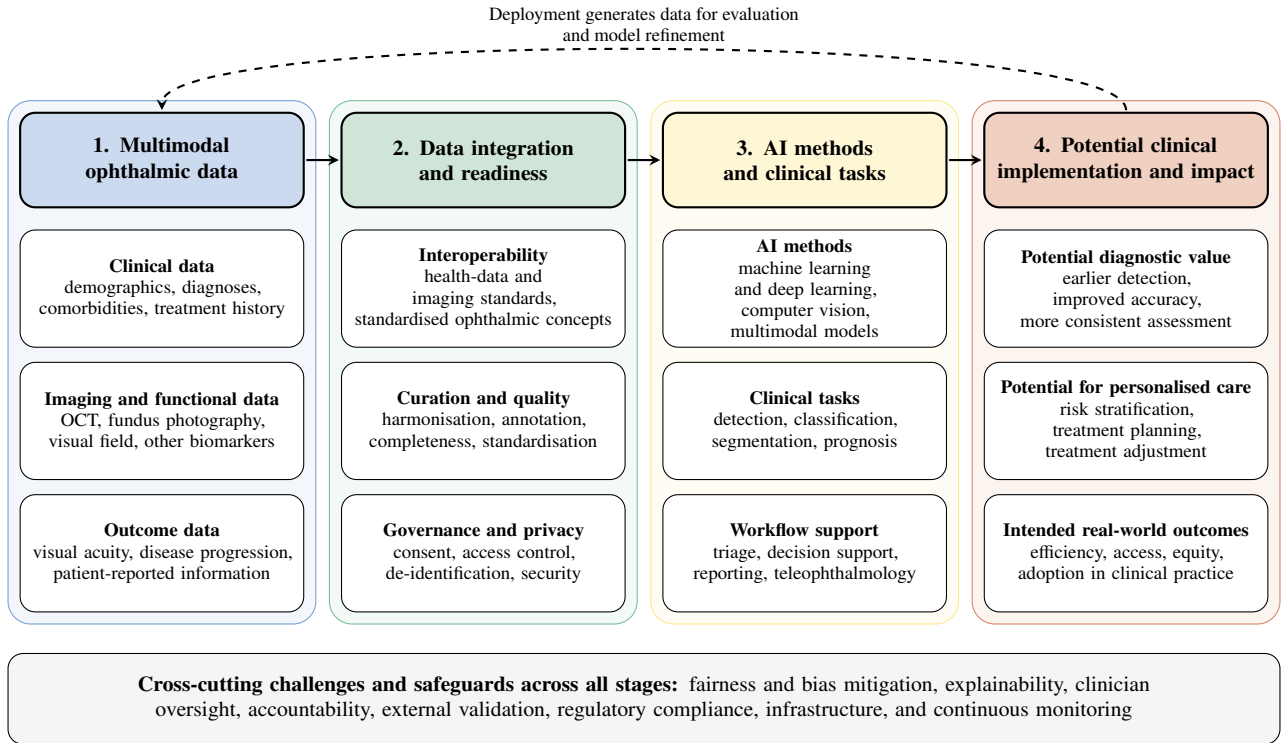


Figure 1: Conceptual pathway from multimodal ophthalmic data to responsible clinical applications of AI. Data integration and readiness precede model development and AI-enabled clinical tasks. Translation into clinical practice also requires privacy protection, fairness, explainability, regulatory compliance, adequate infrastructure, external validation, continuous monitoring, and clinician oversight.

3. Scope of the ophthalmic AI evidence base

Database export and harmonisation yielded 6,482 records for screening and eligibility assessment (Fig. 2). Preprocessing removed 1,460 records (39 duplicates and 1,421 non-English publications), leaving 5,022 for title and abstract screening. Of these, 4,552 did not meet the relevant-task criterion, and the remaining 470 reports were retrieved and assessed for eligibility. A further 43 were excluded on the same criterion, leaving 427 publications for synthesis.

Geographical and co-authorship analyses (Fig. 3) showed that publication activity as well as collaboration were concentrated in relatively few countries, particularly the United States, China, and India, while parts of Africa and South America were comparatively under-represented. Differences in AI research capacity, access to imaging data, screening infrastructure, publication volume, and early deployment might contribute to this pattern, although we did not formally examine these factors. Co-authorship geography alone cannot establish which populations are represented in the underlying datasets. Broader international collaboration spanning populations, devices, disease presentations, care pathways, and resource settings could strengthen equity and external validity.

The literature extends beyond proof-of-concept classification to segmentation, biomarker extraction, prognostic modelling, multimodal integration, screening, and workflow applications. A keyword co-occurrence analysis (Fig. 3) identified four broad domains spanning AI methods, OCT-centred retinal imaging and biomarker analysis, disease diagnosis, monitoring and pro-

gnostic modelling, and translation into screening and clinical workflows. Across these domains, quantitative, multimodal, longitudinal, and decision-support applications can inform clinical decisions on referral, review urgency, monitoring, and follow-up. Their relevance to clinical care is reinforced by the central role of OCT biomarkers, repeated measurements, screening pathways, and treatment decisions in ophthalmology. Assessing technical maturity therefore requires evidence not only on model performance but also on implementation, governance, regulation, cybersecurity, fairness, and post-deployment performance.

Benchmarking for retinal disease classification, segmentation, vascular analysis, and image-quality tasks relied mainly on a small number of reusable public resources, including the Ker-
man dataset, RETOUCH, the Duke spectral-domain OCT (SD-OCT) dataset, OCTA-500, and ROSE-1.^{23–27} These resources support comparison, reproducibility, and technical development, but performance on established benchmarks does not establish generalisability. Further evaluation should be complemented by multi-device, multi-site, longitudinal, and clinically annotated datasets. The supplementary material consolidates these identifiable public datasets and benchmarks.

4. Evidence gaps in clinical translation

We classified the 427 publications across nine general dimensions adapted from recent literature.¹¹ These comprised model performance, technological limitations, real-world impact, data-related factors encompassing availability, integration, quality

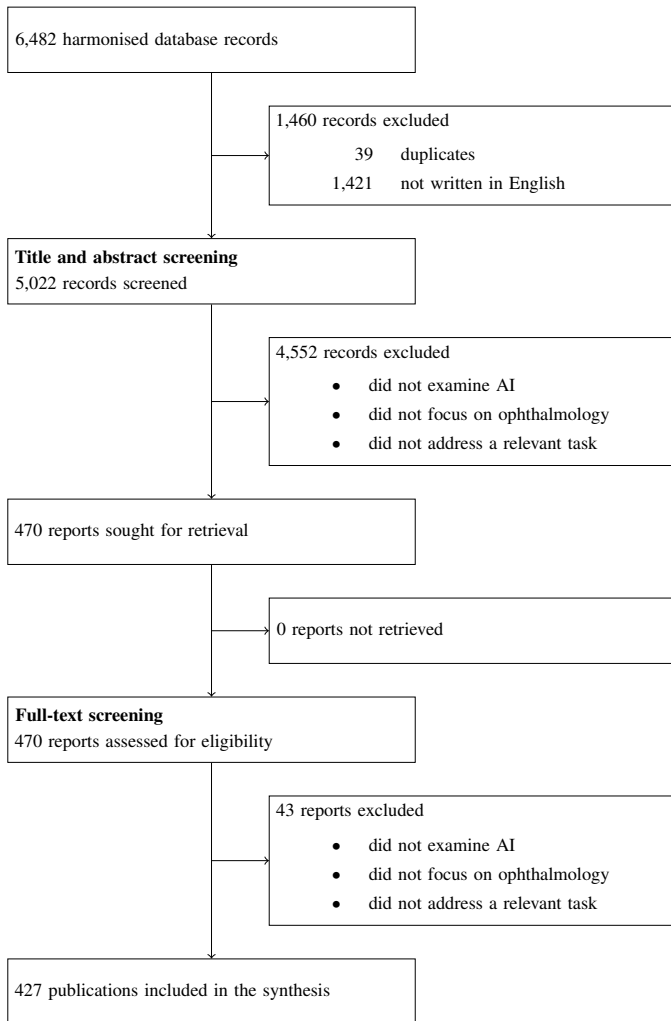


Figure 2: PRISMA 2020 flow diagram of record identification and screening, report retrieval and eligibility assessment, and publication inclusion in this review and evidence gap map of AI in ophthalmology. Records from PubMed, ACM Digital Library, IEEE Xplore, and Scopus were assessed against prespecified criteria during title and abstract screening, followed by full-text screening of retrieved reports. The final synthesis included 427 publications.

and standardisation, infrastructural demands, financial costs, regulatory requirements, cybersecurity risks, and ethical concerns (Fig. 4). The classification yielded 1,287 positive, negative, or balanced assignments based on each publication’s explicit framing rather than inferred or established effects.

Model performance dominated the evidence map, with 235 positive, no negative, and 178 balanced assignments across 416 publications. This pattern reflects a strong emphasis on benchmark evaluation rather than evidence of improved care, whereas technological and data-related dimensions showed more mixed findings. Technological limitations comprised 18 positive, 40 negative, and 276 balanced assignments. Balanced evidence typically paired a failure mode, robustness constraint, generalisation limitation, artefact, or computational restriction with an explicit mitigation strategy, while positive or negative assignments required qualifying evidence in only one direction. Data availability, integration, quality, and standardisation com-

prised 22 positive, 46 negative, and 248 balanced assignments. Evidence in this dimension concerned data scarcity, representativeness, annotation, quality, harmonisation, or integration and corresponding mitigation strategies, rather than routine dataset use.^{28,29}

Nineteen publications received a real-world impact assignment, with 16 positive, one negative, and two balanced. These assignments included reported deployments, clinical or research use, measured patient, clinician, referral, treatment, workflow, or service consequences, and secondary accounts of AI use. A balanced assignment reflected both screening benefits and barriers to referral and treatment.³⁰ All 19 also received model-performance assignments. The category included research-workflow evaluations of human segmentation and correction tasks, and does not represent 19 independent demonstrations of improved care.

Evidence across other implementation dimensions was also sparse. Infrastructural demands received 59 positive, 16 negative, and 31 balanced assignments. Financial costs received 46 positive, five negative, and 19 balanced assignments, excluding computational costs alone. Regulatory requirements received four positive, seven negative, and three balanced assignments, while cybersecurity risks received two positive, two negative, and one balanced assignment. Ten assignments in the general map captured cross-cutting ethical concerns, with specific ethical dimensions assessed separately below. Although infrastructural and financial factors were not uniformly framed as barriers, approval, liability, post-market surveillance, resilience to hostile attack, reimbursement, and formal cost-effectiveness remain largely outside the evaluated literature.

4.1. Ethical dimensions and governance implications

We mapped publication-level framing across nine ethical dimensions adapted from recent literature, including accountability, autonomy, equity, integrity, non-maleficence, privacy, security, transparency, and trust (Fig. 4).¹⁷ This yielded 309 assignments categorised as *caused*, *resolved*, or *caused and resolved*, reflecting how publications framed the relationship between AI and ethical issues rather than establishing ethical effects.

Transparency was the most represented ethical dimension, with 17 caused, 71 resolved, and 53 caused-and-resolved assignments. Mitigations included attribution and feature-contribution outputs, uncertainty estimates, and related explainability tools explicitly linked to interpreting AI decisions,^{31–33} while explainability visualisations without an interpretive claim were insufficient. Privacy received four caused, two resolved, and two caused-and-resolved assignments. Routine consent, ethics approval, anonymisation, de-identification, and restricted access were not considered evidence of AI-mediated privacy protection. Retained reports identified AI-related privacy risks and data-sharing constraints,^{34,35} while mitigation approaches included federated learning without raw-data exchange. However, post-deployment privacy outcomes were not evaluated.^{36,37}

Trust received seven caused, 20 resolved, and 11 caused-and-resolved assignments. Model performance alone was not considered evidence of trust. Reports linked explainability to clinician confidence while also identifying limitations in the

trustworthiness of AI systems.^{32,36} Non-maleficence comprised seven caused, 19 resolved, and eight caused-and-resolved assignments, each explicitly linked to harm or its prevention. Security received two caused, two resolved, and one caused-and-resolved assignment, while integrity received none. Neither security nor integrity was assigned based on model robustness, validation, or data cleaning alone.

Accountability, equity, and autonomy received 16, 55, and 12 assignments, respectively, based on explicit evidence concerning responsibility, liability, subgroup fairness, population representativeness, agency, or meaningful human control. One source described responsibility for AI-driven diagnostic errors as unclear and called for frameworks defining clinical accountability.³⁴ Overall, explanations received considerably more attention than responsibility, meaningful human control, security, or privacy in clinical settings. Addressing these gaps requires greater attention to ethical, regulatory, and governance requirements in study design, validation, deployment planning, and post-deployment monitoring.^{11,17}

5. Discussion

The literature spans diagnostic, predictive, and workflow applications, but evidence mapping revealed substantially greater attention to model performance than to implementation and its consequences. Limited evidence on economics, accountability, regulation, cybersecurity, and privacy further constrains assessment of clinical readiness. These findings highlight the need to determine whether, and under what conditions, technical performance translates into clinical value.

5.1. Data integration and interoperability in clinical translation

Data infrastructure can enable natural language processing (NLP), multimodal modelling, and longitudinal prediction to be integrated into ophthalmic workflows. Broader evaluation across representative sites, devices, populations, and time periods can strengthen generalisability, with external glaucoma studies demonstrating the feasibility of cross-site validation while identifying variation across settings.³⁸ Synthetic data generation, privacy-preserving analysis, structured data linkage, cross-device harmonisation, and EHR integration offer complementary approaches to expanding and connecting available data.^{39–42} Together with representative real-world validation, these approaches could support more robust and transferable ophthalmic AI.

Interoperability can connect clinically relevant information across EHRs, clinical systems, imaging devices, analysis software, and reporting environments. Cross-device registration can align complementary OCT signals across imaging systems.⁴¹ Standards such as Fast Healthcare Interoperability Resources (FHIR) and Digital Imaging and Communications in Medicine (DICOM) support syntactic interoperability and data exchange,^{43,44} while semantic interoperability requires consistent representation of ophthalmic information. Existing terminologies, including the International Classification of Diseases (ICD), the International Classification of Health Interventions (ICHI),

and SNOMED Clinical Terms (SNOMED CT), can be complemented by representations of imaging-derived biomarkers, treatment histories, and longitudinal disease states.^{45,46} Mapping interfaces, metadata, and data-quality constraints, together with shared outcome definitions and machine-readable ophthalmic concepts, could support registry-scale validation and multi-site workflow integration.

Data quality influences model development, robustness, behaviour, and clinical usefulness through annotation reliability, completeness, expert adjudication, image quality, disease-label definitions, and clinical context.^{28,29} Structured assessment frameworks and standardised annotation practices can strengthen the consistency, transparency, and reproducibility of data-quality evaluation.^{47,48} Comprehensive dataset documentation should report image or scan counts, acquisition context, device mix, disease spectrum, label source, missingness, longitudinal follow-up, treatment history, and intended use, helping clinicians and regulators assess whether findings reflect transferable clinical information or dataset-specific signals.⁴⁹

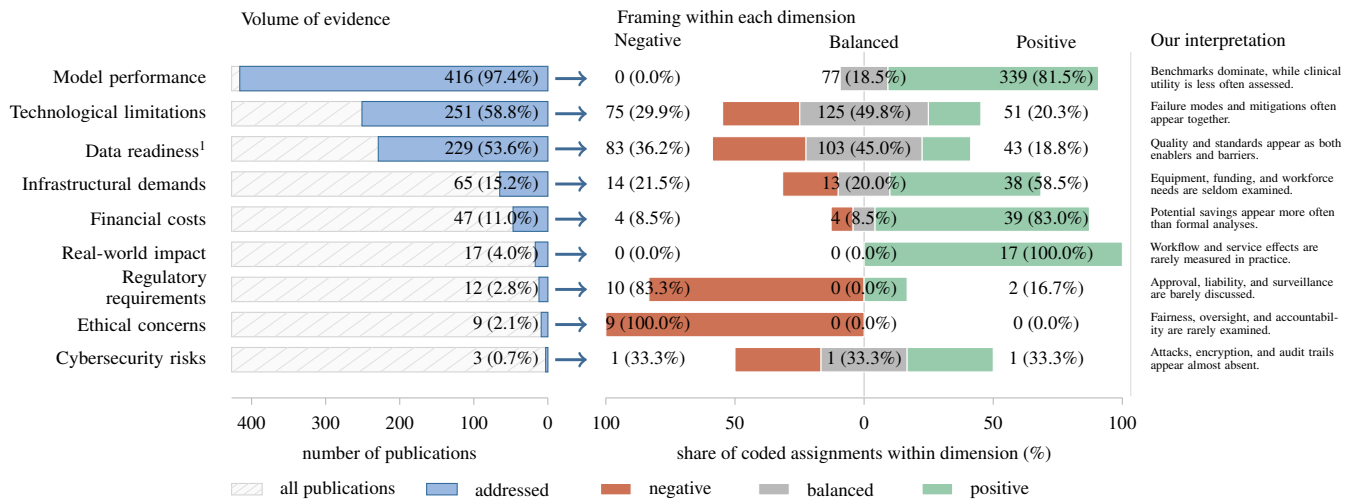
Retinal imaging poses specific privacy risks because metadata removal and pseudonymisation do not preclude re-identification, and patient-specific vascular and anatomical features can enable biometric matching.^{50,51} Evidence is strongest for fundus vascular patterns and does not establish comparable identifiability across all OCT data. Retinal and OCT datasets should therefore be treated as potentially identifiable health data and protected through data minimisation, purpose limitation, access control, encryption, secure linkage, re-identification testing, and explicit governance of secondary use. Federated learning and other privacy-preserving approaches can limit the exchange of raw data but do not eliminate privacy risk.^{40,52}

5.2. Clinical translation requires diagnostic, predictive, and workflow support

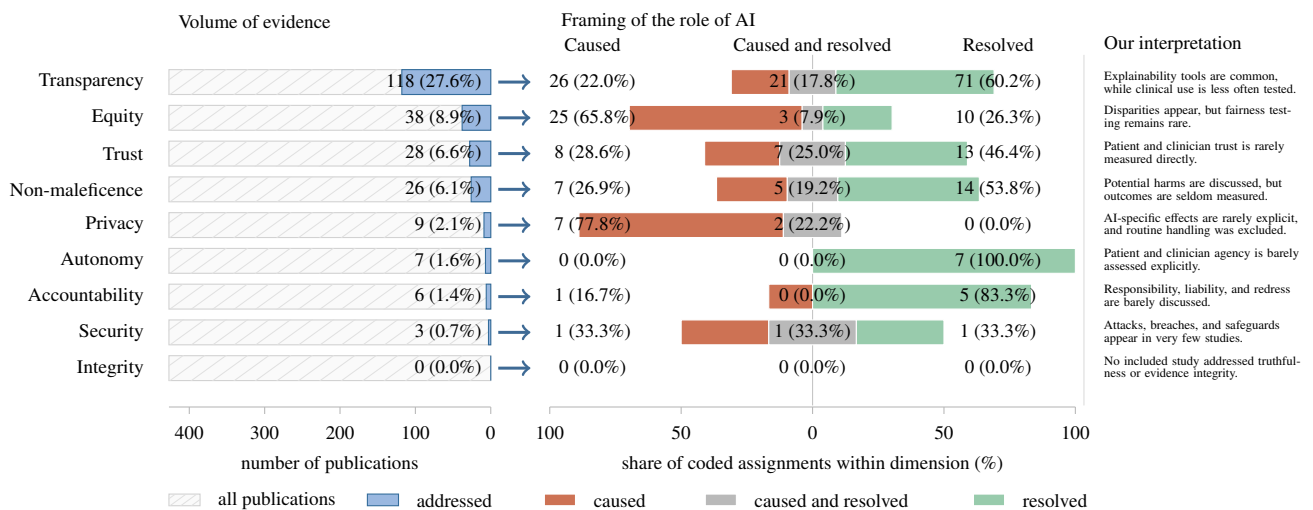
Diagnostic AI is among the most established applications of AI in ophthalmology, with OCT supporting detection, classification, segmentation, biomarker quantification, monitoring, and treatment planning. Deep-learning systems have achieved strong performance in retinal disease detection and quantification, including geographic atrophy assessment,^{7,53,54} and have approached or exceeded expert performance on selected benchmarks.^{8,23,55} In DR screening, teleophthalmology platforms could translate these capabilities into earlier detection, greater access, and more standardised care.^{56–58} Clearly defined tasks, intended uses, and clinical workflows could further enable such systems to extend specialist capacity and improve the delivery of eye care.

Combining OCT with fundus photography, OCTA, or structured clinical variables might better reflect the information available for clinical decision-making, provided that harmonisation, missing-data handling, validation, interpretability, as well as workflow integration are addressed.^{42,59,60} Reported productivity gains are encouraging,⁶¹ but evaluation should extend beyond efficiency to patient and service outcomes, including timely referral, diagnostic consistency, treatment timing, avoidable visits, workload, safety, equity, and cost-effectiveness. Clinical value ultimately depends on whether model outputs improve decisions

Evidence gap maps for general and ethical dimensions of AI in ophthalmology (427 publications)



(A) Evidence volume and dominant framing across general dimensions



(B) Evidence volume and framing of artificial intelligence across ethical dimensions

Figure 4: Evidence gap maps of thematic coverage and framing across the 427 included publications on AI in ophthalmology. Dimensions are ordered by decreasing evidence volume within each panel. (A) The left-hand bars show the number of publications addressing each general dimension, while the diverging bars show whether implications were coded as negative, balanced, or positive. Performance reporting was common, but real-world effects, regulatory requirements, ethical concerns, and cybersecurity received substantially less attention. (B) The left-hand bars show publication counts for ethical dimensions, while the diverging bars show whether AI was framed as causing the issue, resolving it, or both. Transparency predominated, whereas evidence on equity, trust, autonomy, accountability, security, and integrity was limited. The neutral grey segment in each diverging bar is centred on zero and divided equally between the two sides. Percentages were calculated within each row. Publications could be assigned to multiple dimensions, so totals across rows and categories exceed 427.

and care rather than simply adding information to existing workflows.

Predictive modelling extends AI from characterising current disease to anticipating its course through estimates of visual acuity, treatment response, recurrence, progression, or retreatment burden. These predictions could inform earlier intervention, individualised follow-up, resource allocation, and shared decision-making.^{9,10,62–64} Approaches range from regression-based models to deep-learning systems that distinguish progression trajectories.^{42,65} By integrating OCT biomarkers, treatment history, longitudinal outcomes, and clinical metadata, predictive models can capture disease trajectories and the temporal context of care beyond single-image classification. Their clinical value, however, depends on robust validation across populations and settings and on predictions that are interpretable and actionable.

Predictive models could identify patients at high risk of progression, supporting tailored monitoring and timely treatment adjustment. However, performance and calibration in narrow retrospective cohorts may not translate into clinical utility in broader practice. Evaluation should therefore account for adherence, comorbidity, treatment variation, access to care, clinician decisions, disease severity, and social or behavioural factors that are often incompletely captured. Beyond calibration, uncertainty communication, interpretability, and temporal robustness, evaluation should establish whether model outputs are actionable, change clinical decisions, and ultimately improve patient or service outcomes.^{18,66}

5.3. Workflow integration makes real-world value testable

Clinical implementation is necessary to determine whether ophthalmic AI delivers meaningful benefits in practice. Institutional stakeholders prioritise clinical decision support, predictive analytics, and workflow optimisation as opportunities to improve care quality, reduce operational burden, and support resource allocation.⁶⁷ However, regulatory and legal uncertainty, limited expertise, model-safety concerns, and unclear returns on investment continue to impede adoption.⁶⁸ Public acceptance remains limited, with surveyed US adults reporting more discomfort than enthusiasm and fewer than half expecting AI-assisted care to improve outcomes.⁶⁹ Figure 5 places these expected benefits and implementation barriers within the staged roadmap for clinical translation.

Reports of AI integration into ophthalmic clinical workflows remain uncommon but provide early evidence of translation beyond retrospective evaluation. Examples include community screening with immediate AI-generated referral recommendations, AI-assisted OCT screening linked to referral and treatment, and a cloud-based OCT platform for retinal-anomaly screening across institutions.^{30,60,70} Together, these reports demonstrate implementation across defined tasks, populations, and care settings. Broader adoption will require integration into local workflows and clinical information systems, alignment with regulatory pathways, and minimal additional cognitive or administrative burden.

Clinical dashboards, for instance, could integrate OCT biomarkers, visual acuity, treatment history, disease activity, imaging quality, and follow-up intervals into a coherent view of the

patient before predictive modelling is introduced. By connecting fragmented data, they could also reveal data-quality problems, support clinical review, and provide an auditable context for subsequent AI outputs. Ophthalmic AI research should therefore incorporate implementation science to determine whether systems improve clinical decisions, outcomes, access, equity, safety, or efficiency in practice. Evaluation should include prospective validation, human–AI interaction studies, workflow and subgroup assessment, failure-mode monitoring, and health economic analysis, alongside appropriate clinician oversight.^{18,19,58,71}

5.4. Explanations must support effective human oversight

Transparency was the most frequently mapped ethical dimension, but providing explanations does not establish that AI outputs are clinically interpretable. Explanations may help clinicians understand outputs, identify failures, and interpret uncertainty, but saliency maps alone do not demonstrate improved error recognition or decision-making.^{72,73} Evaluation should therefore determine whether explanations improve human–AI decision-making, support recognition of model failures, and communicate uncertainty and limitations effectively.

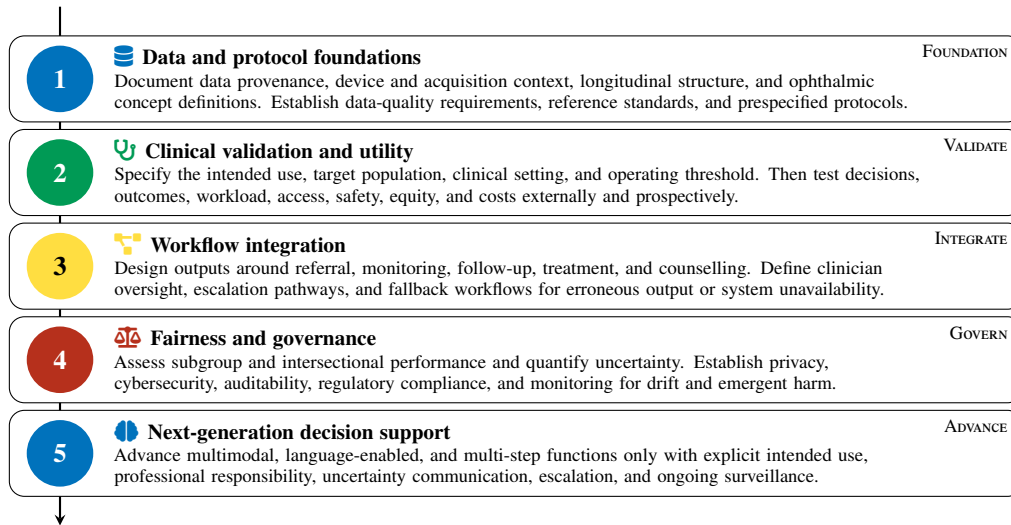
Effective oversight of AI-supported diagnosis or treatment requires clinicians to understand intended-use boundaries, interpret actionable outputs, and challenge, override, or escalate them when necessary. These capabilities support appropriate reliance on AI while preserving meaningful clinician control.^{18,74} Many AI-enabled medical-device applications fall within the high-risk regime of Regulation (EU) 2024/1689, which establishes requirements for human oversight. Article 14 requires human oversight to support understanding of system limitations, avoidance of over-reliance, interpretation of outputs, and the ability to disregard, override, or interrupt the system.⁷⁵

It distributes obligations across providers and deployers, enabling responsibilities to be allocated without treating AI systems as responsible actors or placing all obligations on treating physicians. Institutions and professionals also retain duties arising from clinical practice and other applicable law. Before deployment, governance should define responsibility for accepting or overriding outputs, escalation, monitoring, and incident response. Clinical services should maintain staffed fallback procedures for erroneous outputs, performance degradation, cyber incidents, or downtime while preserving clinicians' diagnostic competence. These safeguards can strengthen resilience, limit automation bias and deskilling, and maintain the expertise needed to recognise and respond to system failures.

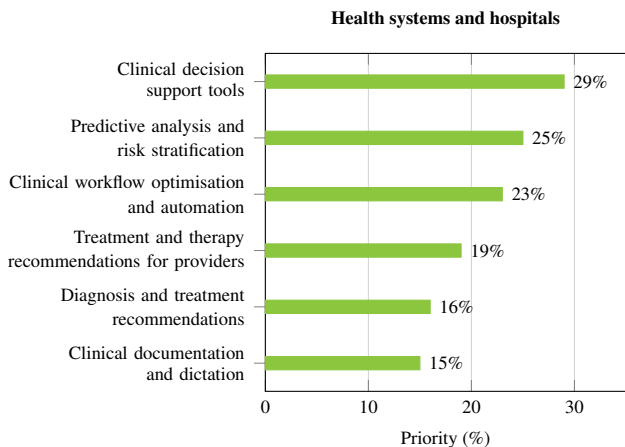
5.5. From technical performance to accountable clinical value

Future evaluation should define the clinical decision AI is intended to support and determine where it could add value to the existing care pathway. Following external validation across representative populations, devices, and settings, prospective studies should examine whether AI supports more timely referral and treatment, more individualised monitoring, greater efficiency, or improved access to care. Evaluation should consider these potential benefits alongside uncertainty, subgroup performance, safety, and costs, while post-deployment monitoring can help sustain performance and clinical value over time.

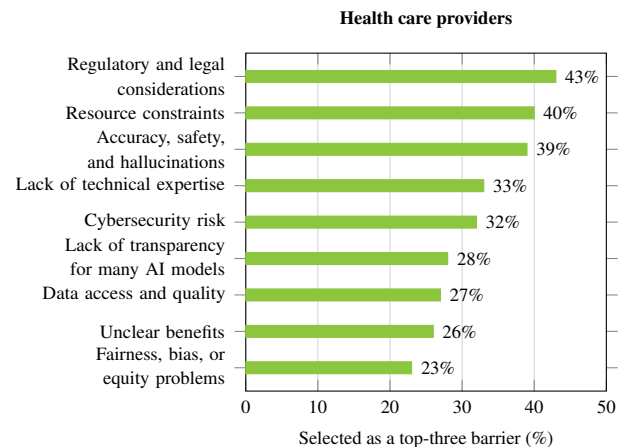
Move beyond benchmarks: demonstrate the clinical value of ophthalmic AI



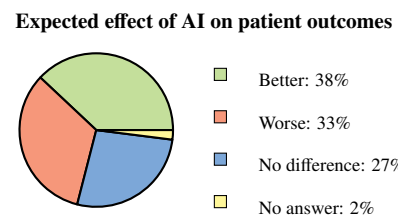
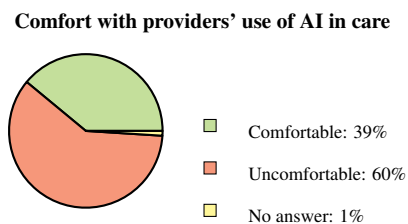
(A) Translational roadmap



(B) Priority use cases



(C) Implementation barriers



(D) Public attitudes

Figure 5: Translation priorities and data from selected surveys and reports on clinical AI. (A) Five connected stages link data and protocol foundations to clinical validation and utility, workflow integration, fairness and governance, and next-generation decision support. Benchmark measures of discrimination or accuracy are insufficient on their own. Technical performance must be translated into prospective evidence of equitable and accountable clinical value. (B) Health systems and hospitals most often identified clinical decision support, predictive analysis and risk stratification, and workflow optimisation as priority use cases.⁶⁷ (C) Health care providers most often ranked regulatory and legal considerations, resource constraints, and concerns about accuracy, safety, and hallucinations among the leading barriers to adoption.⁶⁸ (D) Fewer than half of adults in the United States reported being comfortable with providers relying on AI in their care, and fewer than half expected AI to improve patient outcomes.⁶⁹ These findings reflect specific geographical and institutional contexts and are not globally representative.

Multimodal and more autonomous systems could integrate imaging, clinical records, and patient-reported information to support longitudinal decision-making and follow-up planning, but their value depends on whether outputs can inform appropriate clinical action. Development should therefore combine model evaluation with interoperable data, privacy and security safeguards, workflow integration, meaningful human oversight, and clear responsibility for decisions and escalation. For systems performing linked tasks, evaluation should examine how errors propagate across tasks and whether clinicians can recognise failures, intervene effectively, and restore safe care.

6. Strengths and limitations

This review combines structured searching, thematic analysis, keyword co-occurrence mapping, international co-authorship analysis, public-dataset mapping, and evidence gap mapping to characterise the technical scope, geographical distribution, reusable resources, and translational gaps in ophthalmic AI. Publication-level mappings and supplementary dataset documentation enhance the traceability of the synthesis through evidence locators and rationales for retained assignments.

Several limitations affect interpretation. First, although no lower publication-date limit was prespecified, the eligible literature spanned January 2010 to April 2026, with the starting point reflecting retrieval rather than a predefined restriction. Later publications could alter the evidence distribution. Second, requiring OCT, OCTA, or related ophthalmic imaging limited coverage of non-imaging applications. Third, restricting eligibility to English might have excluded relevant work. Fourth, heterogeneity in publication type, clinical indication, AI task, imaging modality, dataset source, validation strategy, and outcome reporting precluded meta-analysis. Fifth, the maps capture publication-level framing rather than effect sizes, diagnostic accuracy, risk of bias, or certainty ratings. Sixth, assignments remain interpretive despite explicit coding rules and an audit trail, and we did not assess study-level risk of bias. The audit strengthens traceability, but the counts do not establish efficacy. Finally, limited reporting of demographic and socioeconomic characteristics constrained assessment of population coverage and equity.^{15,16}

7. Conclusion

Ophthalmic AI extends from image interpretation to prediction and workflow support, but evidence of clinical value has not kept pace with its expanding applications. This review identified priorities for evaluation while recognising that limited evidence does not imply lack of benefit. The central question is whether model outputs inform actionable clinical decisions and whether the resulting care improves outcomes for patients or health services.

Progress therefore depends on evaluating AI within its intended care pathway, using representative populations, credible comparators, effective human oversight, and measures of safety, equity, and cost. Interoperable data and clear responsibility are integral to this process because they shape how AI outputs are

interpreted and acted on. Ophthalmology's repeated imaging, longitudinal follow-up, and established care pathways make it particularly well suited to determining whether advances in AI translate into meaningful clinical value.

Contributors

T.S., R.V., and M.W. conceptualised the review. T.S. and M.F. developed the methodology. T.S. and R.V. developed the software and data-processing workflow, provided resources, and curated the publication-level data. T.S. and M.F. performed the validation and formal analysis. T.S., M.F., and R.V. screened records, assessed eligibility, extracted data, and interpreted the findings. T.S. created the original visualisations, and B.P. contributed visualisation concepts and figure feedback, including the evidence-map presentation. B.P., M.V., D.K., M.R., M.G., and M.W. provided substantive manuscript feedback, interpretive guidance, and critical review of the structure, methods, results, and discussion. T.S., M.F., R.V., and M.W. drafted the original manuscript, and all authors critically reviewed it. T.S., M.F., and R.V. had full access to the publication-level data underlying the review and verified the data. All authors reviewed and approved the final version and take responsibility for the manuscript content and the decision to submit.

Declaration of interests

We declare no competing interests.

Data sharing

Publication-level extraction tables, evidence gap maps, coding rules, and public-dataset documentation are provided in the supplementary material. An interactive evidence explorer is available at <https://tschlosser13.github.io/ophthalmic-ai-evidence-portal/>. No individual participant data were collected or analysed. Additional analysis code and extraction files are available from the corresponding author on reasonable request.

Declaration of AI use

During the preparation of this work, the authors used OpenAI Codex (GPT-5.6) to support language editing, improve clarity and consistency, and assist with the preparation of supplementary material. The authors reviewed and edited all AI-assisted output and take full responsibility for the content of the publication.

Role of the funding source

No specific funding was received for this review. No funder had any role in the review design, the collection, analysis, or interpretation of the literature, the writing of the manuscript, or the decision to submit it for publication.

References

- [1] M. M. Brown, G. C. Brown, S. Sharma, J. Kistler, H. Brown, Utility values associated with blindness in an adult population, *British Journal of Ophthalmology* 85 (3) (2001) 327–331. doi:10.1136/bjo.85.3.327. URL <https://doi.org/10.1136/bjo.85.3.327>
- [2] J. G. Ravin, Sesquicentennial of the ophthalmoscope, *Archives of Ophthalmology* 117 (12) (1999) 1634–1638. doi:10.1001/archophth.117.12.1634. URL <https://doi.org/10.1001/archophth.117.12.1634>
- [3] D. Huang, E. A. Swanson, C. P. Lin, J. S. Schuman, W. G. Stinson, W. Chang, M. R. Hee, T. Flotte, K. Gregory, C. A. Puliafito, J. G. Fujimoto, Optical coherence tomography, *Science* 254 (5035) (1991) 1178–1181. doi:10.1126/science.1957169. URL <https://doi.org/10.1126/science.1957169>
- [4] Y. Jia, O. Tan, J. Tokayer, B. Potsaid, Y. Wang, J. J. Liu, M. F. Kraus, H. Subhash, J. G. Fujimoto, J. Hornegger, D. Huang, Split-spectrum amplitude-decorrelation angiography with optical coherence tomography, *Optics Express* 20 (4) (2012) 4710–4725. doi:10.1364/OE.20.004710. URL <https://doi.org/10.1364/OE.20.004710>
- [5] GBD 2019 Blindness and Vision Impairment Collaborators, Vision Loss Expert Group of the Global Burden of Disease Study, Causes of blindness and vision impairment in 2020 and trends over 30 years, and prevalence of avoidable blindness in relation to VISION 2020: The right to sight: An analysis for the global burden of disease study, *The Lancet Global Health* 9 (2) (2021) e144–e160. doi:10.1016/S2214-109X(20)30489-7. URL [https://doi.org/10.1016/S2214-109X\(20\)30489-7](https://doi.org/10.1016/S2214-109X(20)30489-7)
- [6] P. Song, Y. Xu, M. Zha, Y. Zhang, I. Rudan, Global epidemiology of retinal vein occlusion: A systematic review and meta-analysis of prevalence, incidence, and risk factors, *Journal of Global Health* 9 (1) (2019) 010427. doi:10.7189/jogh.09.010427. URL <https://doi.org/10.7189/jogh.09.010427>
- [7] U. Schmidt-Erfurth, A. Sadeghipour, B. S. Gerendas, S. M. Waldstein, H. Bogunović, Artificial intelligence in retina, *Progress in Retinal and Eye Research* 67 (2018) 1–29. doi:10.1016/j.preteyeres.2018.07.004. URL <https://doi.org/10.1016/j.preteyeres.2018.07.004>
- [8] J. D. Fauw, J. R. Ledsam, B. Romera-Paredes, S. Nikolov, N. Tomasev, S. Blackwell, H. Askham, X. Glorot, B. O'Donoghue, D. Visentin, G. v. d. Driessche, B. Lakshminarayanan, C. Meyer, F. Mackinder, S. Bouton, K. Ayoub, R. Chopra, D. King, A. Karthikesalingam, C. O. Hughes, R. Raine, J. Hughes, D. A. Sim, C. Egan, A. Tufail, H. Montgomery, D. Hassabis, G. Rees, T. Back, P. T. Khaw, M. Suleyman, J. Cornebise, P. A. Keane, O. Ronneberger, Clinically applicable deep learning for diagnosis and referral in retinal disease, *Nature Medicine* 24 (9) (2018) 1342–1350. doi:10.1038/s41591-018-0107-6. URL <https://www.nature.com/articles/s41591-018-0107-6>
- [9] J. Yim, R. Chopra, T. Spitz, J. Winkens, A. Obika, C. Kelly, H. Askham, M. Lukic, J. Huemer, K. Fasler, G. Moraes, C. Meyer, M. Wilson, J. Dixon, C. Hughes, G. Rees, P. T. Khaw, A. Karthikesalingam, D. King, D. Hassabis, M. Suleyman, T. Back, J. R. Ledsam, P. A. Keane, J. De Fauw, Predicting conversion to wet age-related macular degeneration using deep learning, *Nature Medicine* 26 (6) (2020) 892–899. doi:10.1038/s41591-020-0867-7. URL <https://doi.org/10.1038/s41591-020-0867-7>
- [10] H. Bogunović, V. Mares, G. S. Reiter, U. Schmidt-Erfurth, B. H. M. V. R. GS, S.-E. U. R. G.S., Predicting treat-and-extend outcomes and treatment intervals in neovascular age-related macular degeneration from retinal optical coherence tomography using artificial intelligence, *Frontiers in medicine* 9 (2022) 958469. doi:10.3389/fmed.2022.958469. URL <https://pubmed.ncbi.nlm.nih.gov/36017006/>
- [11] T. F. Tan, A. J. Thirunavukarasu, L. Jin, J. Lim, S. Poh, Z. L. Teo, M. Ang, R. V. P. Chan, J. Ong, A. Turner, J. Karlström, T. Y. Wong, J. Stern, D. S.-W. Ting, Artificial intelligence and digital health in global eye health: Opportunities and challenges, *The Lancet Global Health* 11 (9) (2023) e1432–e1443. doi:10.1016/S2214-109X(23)00323-6. URL [https://doi.org/10.1016/S2214-109X\(23\)00323-6](https://doi.org/10.1016/S2214-109X(23)00323-6)
- [12] Y. Liu, N. M. Holekamp, J. S. Heier, Prospective, longitudinal study: Daily self-imaging with home OCT for neovascular age-related macular degeneration, *Ophthalmology Retina* 6 (7) (2022) 575–585. doi:10.1016/j.oret.2022.02.011. URL <https://doi.org/10.1016/j.oret.2022.02.011>
- [13] A. J. Thirunavukarasu, S. Mahmood, A. Malem, W. P. Foster, R. Sanghera, R. Hassan, S. Zhou, S. W. Wong, Y. L. Wong, Y. J. Chong, A. Shakeel, Y.-H. Chang, B. K. J. Tan, N. Jain, T. F. Tan, S. Rauz, D. S. W. Ting, D. S. J. Ting, Large language models approach expert-level clinical knowledge and reasoning in ophthalmology: A head-to-head cross-sectional study, *PLOS Digital Health* 3 (4) (2024) e0000341. doi:10.1371/journal.pdig.0000341. URL <https://doi.org/10.1371/journal.pdig.0000341>
- [14] S. Y. Wang, B. Tseng, T. Hernandez-Boussard, Development and evaluation of novel ophthalmology domain-specific neural word embeddings to predict visual prognosis, *International Journal of Medical Informatics* 150 (2021). doi:10.1016/j.ijmedinf.2021.104464. URL <https://www.sciencedirect.com/science/article/abs/pii/S1386505621000903>
- [15] C. Lam, Y. L. Wong, Z. Tang, X. Hu, T. X. Nguyen, D. Yang, S. Zhang, J. Ding, S. K. H. Szeto, A. R. Ran, C. Y. Cheung, L. C. W. YL, T. Z. H. X, N. TX, Y. D. Z. S. D. J, S. SKH, R. AR, C. CY, W. YL., N. TX., S. S.K.H., R. A.R., C. C.Y., Performance of artificial intelligence in detecting diabetic macular edema from fundus photography and optical coherence tomography images: A systematic review and meta-analysis, *Diabetes care* 47 (2) (2024) 304–319. doi:10.2337/dc23-0993. URL <https://pubmed.ncbi.nlm.nih.gov/38241500/>
- [16] S. M. Khan, X. Liu, S. Nath, E. Korot, L. Faes, S. K. Wagner, P. A. Keane, N. J. Sebire, M. J. Burton, A. K. Denniston, A global review of publicly available datasets for ophthalmological imaging: barriers to access, usability, and generalisability, *The Lancet Digital Health* 3 (1) (2021) e51–e66. doi:10.1016/S2589-7500(20)30240-5. URL [https://doi.org/10.1016/S2589-7500\(20\)30240-5](https://doi.org/10.1016/S2589-7500(20)30240-5)
- [17] Y. Ning, S. Teixayavong, Y. Shang, J. Savulescu, V. Nagaraj, D. Miao, M. Mertens, D. S. W. Ting, J. C. L. Ong, M. Liu, J. Cao, M. Dunn, R. Vaughan, M. E. H. Ong, J. J.-Y. Sung, E. J. Topol, N. Liu, Generative artificial intelligence and ethical considerations in health care: A scoping review and ethics checklist, *The Lancet Digital Health* 6 (11) (2024) e848–e856. doi:10.1016/S2589-7500(24)00143-2. URL [https://doi.org/10.1016/S2589-7500\(24\)00143-2](https://doi.org/10.1016/S2589-7500(24)00143-2)
- [18] B. Vasey, M. Nagendran, B. Campbell, D. A. Clifton, G. S. Collins, S. Denaxas, A. K. Denniston, L. Faes, B. Geerts, M. Ibrahim, X. Liu, B. A. Mateen, P. Mathur, M. D. McCradden, M. Morgan, J. Ordish, C. Rogers, S. Saria, D. S. W. Ting, P. Watkinson, W. Weber, P. Wheatstone, P. McCulloch, DECIDE-AI expert group, Reporting guideline for the early-stage clinical evaluation of decision support systems driven by artificial intelligence: DECIDE-AI, *Nature Medicine* 28 (5) (2022) 924–933. doi:10.1038/s41591-022-01772-9. URL <https://doi.org/10.1038/s41591-022-01772-9>
- [19] X. Liu, S. Cruz Rivera, D. Moher, M. J. Calvert, A. K. Denniston, SPIRIT-AI and CONSORT-AI Working Group, Reporting guidelines for clinical trial reports for interventions involving artificial intelligence: the CONSORT-AI extension, *Nature Medicine* 26 (9) (2020) 1364–1374. doi:10.1038/s41591-020-1034-x. URL <https://doi.org/10.1038/s41591-020-1034-x>
- [20] R. M. Wolf, R. Channa, T. Y. A. Liu, A. Zehra, L. Bromberger, D. Patel, A. Ananthakrishnan, E. A. Brown, L. Prichett, H. P. Lehmann, M. D. Abramoff, Autonomous artificial intelligence increases screening and follow-up

- for diabetic retinopathy in youth: the ACCESS randomized control trial, *Nature Communications* 15 (1) (2024) 421. doi:10.1038/s41467-023-44676-z. URL <https://doi.org/10.1038/s41467-023-44676-z>
- [21] M. J. Page, J. E. McKenzie, P. M. Bossuyt, I. Boutron, T. C. Hoffmann, C. D. Mulrow, L. Shamseer, J. M. Tetzlaff, E. A. Akl, S. E. Brennan, R. Chou, J. Glanville, J. M. Grimshaw, A. Hróbjartsson, M. M. Lalu, T. Li, E. W. Loder, E. Mayo-Wilson, S. McDonald, L. A. McGuinness, L. A. Stewart, J. Thomas, A. C. Tricco, V. A. Welch, P. Whiting, D. Moher, The PRISMA 2020 statement: An updated guideline for reporting systematic reviews, *BMJ* 372 (2021) n71. doi:10.1136/bmj.n71. URL <https://doi.org/10.1136/bmj.n71>
- [22] H. White, B. Albers, M. Gaarder, H. Kornør, J. Littell, Z. Marshall, C. Mathew, T. Pigott, B. Snilstveit, H. Waddington, V. Welch, Guidance for producing a Campbell evidence and gap map, *Campbell Systematic Reviews* 16 (4) (2020) e1125. doi:10.1002/c12.1125. URL <https://doi.org/10.1002/c12.1125>
- [23] D. S. Kermany, M. Goldbaum, W. Cai, C. C. S. Valentim, H. Liang, S. L. Baxter, A. McKeown, G. Yang, X. Wu, F. Yan, J. Dong, M. K. Prasadha, J. Pei, M. Y. L. Ting, J. Zhu, C. Li, S. Hewett, J. Dong, I. Ziyar, A. Shi, R. Zhang, L. Zheng, R. Hou, W. Shi, X. Fu, Y. Duan, V. A. N. Huu, C. Wen, E. D. Zhang, C. L. Zhang, O. Li, X. Wang, M. A. Singer, X. Sun, J. Xu, A. Tafreshi, M. A. Lewis, H. Xia, K. Zhang, Identifying medical diagnoses and treatable diseases by image-based deep learning, *Cell* 172 (5) (2018) 1122–1131.e9. doi:10.1016/j.cell.2018.02.010. URL <https://doi.org/10.1016/j.cell.2018.02.010>
- [24] H. Bogunović, F. Venhuizen, S. Klimscha, S. Apostolopoulos, A. Bab-Hadiashar, U. Bagci, M. F. Beg, L. Bekalo, Q. Chen, C. Ciller, K. Gopinath, A. K. Gostar, K. Jeon, Z. Ji, S. H. Kang, D. D. Koozekanani, D. Lu, D. Morley, K. K. Parhi, H. S. Park, A. Rashno, M. Sarunic, S. Shaikh, J. Sivaswamy, R. Tennakoon, S. Yadav, S. De Zanet, S. M. Waldstein, B. S. Gerendas, C. I. Sanchez, U. Schmidt-Erfurth, RETOUCH: The retinal OCT fluid detection and segmentation benchmark and challenge, *IEEE Transactions on Medical Imaging* 38 (8) (2019) 1858–1874. doi:10.1109/TMI.2019.2901398. URL <https://doi.org/10.1109/TMI.2019.2901398>
- [25] S. J. Chiu, M. J. Allingham, P. S. Mettu, S. W. Cousins, J. A. Izatt, S. Farsiu, Kernel regression based segmentation of optical coherence tomography images with diabetic macular edema, *Biomedical Optics Express* 6 (4) (2015) 1172–1194. doi:10.1364/B0E.6.001172. URL <https://doi.org/10.1364/B0E.6.001172>
- [26] M. Li, K. Huang, Q. Xu, J. Yang, Y. Zhang, Z. Ji, K. Xie, S. Yuan, Q. Liu, Q. Chen, OCTA-500: A retinal dataset for optical coherence tomography angiography study, *Medical Image Analysis* 93 (2024) 103092. doi:10.1016/j.media.2024.103092. URL <https://doi.org/10.1016/j.media.2024.103092>
- [27] Y. Ma, H. Hao, J. Xie, H. Fu, J. Zhang, J. Yang, Z. Wang, J. Liu, Y. Zheng, Y. Zhao, ROSE: A retinal OCT-angiography vessel segmentation dataset and new model, *IEEE Transactions on Medical Imaging* 40 (3) (2021) 928–939. doi:10.1109/TMI.2020.3042802. URL <https://doi.org/10.1109/TMI.2020.3042802>
- [28] Z. Tang, X. Wang, A. R. Ran, D. Yang, A. Ling, J. C. Yam, X. Zhang, S. K. H. Szeto, J. Chan, C. Y. K. Wong, V. W. K. Hui, C. K. M. Chan, T. Y. Wong, C.-Y. Cheng, C. Sabanayagam, Y. C. Tham, G. Liew, G. Anantharaman, R. Raman, Y. Cai, H. Che, L. Luo, Q. Liu, Y. L. Wong, A. K. Y. Ngai, V. L. Yuen, N. Kei, T. Y. Y. Lai, H. Chen, C. C. Tham, P.-A. Heng, C. Y. Cheung, T. Z. W. X. R. AR, Y. D. L. A. Y. JC, Z. X. S. SKH, C. J. W. CYK, H. VWK, C. CKM, W. TY, C. CY, S. C. T. YC, L. G. A. G. R. R. C. Y. C. H. L. L. Q. W. YL, N. AKY, Y. VL, K. N. L. TYY, C. H. T. CC, H. PA. C. CY, R. A.R., Y. J.C., S. S.K.H., W. C.Y.K., H. V.W.K., C. C.K.M., W. T.Y., C. C.-Y., T. Y.C., W. Y.L., N. A.K.Y., Y. V.L., L. T.Y.Y., T. C.C., H. P.-A., C. C.Y., Deep learning-based image quality assessment for optical coherence tomography macular scans: a multicentre study, *British Journal of Ophthalmology* 108 (11) (2024) 1555–1563. doi:10.1136/bjo-2023-323871. URL <https://pubmed.ncbi.nlm.nih.gov/39033014/>
- [29] K. A. Thakoor, X. Li, E. Tsamis, Z. Z. Zemborain, C. G. De Moraes, P. Sajda, D. C. Hood, T. KA, L. X, T. E, Z. ZZ, D. M. CG, S. P, H. DC, T. K.A., Z. Z.Z., D. M. C.G., H. D.C., Strategies to improve convolutional neural network generalizability and reference standards for glaucoma detection from oct scans, *Translational vision science & technology* 10 (4) (2021) 16. doi:10.1167/tvst.10.4.16. URL <https://pubmed.ncbi.nlm.nih.gov/34003990/>
- [30] D. Liu, X. Xu, J. Zhao, N. Tian, Y. An, X. Liu, J. Yu, X. Yang, T. Niu, L. D., X. X., Z. J., T. N., A. Y., L. X., Y. J., Y. X., N. T., Analysis of early screening results for high-risk populations of retinal diseases based on artificial intelligence optical coherence tomography technology, *Chinese Journal of Ocular Fundus Diseases* 42 (2) (2026) 127–133. doi:10.3760/cma.j.cn511434-20251009-00423. URL <https://www.scopus.com/pages/publications/105031493897?origin=resultslist>
- [31] K. A. Thakoor, S. C. Koorathota, D. C. Hood, P. Sajda, K. A. Thakoor, S. C. Koorathota, D. C. Hood, P. Sajda, T. KA, K. SC, H. DC, S. P, T. K.A., K. S.C., H. D.C., Robust and interpretable convolutional neural networks to detect glaucoma in optical coherence tomography images, *IEEE Transactions on Biomedical Engineering* 68 (8) (2021) 2456–2466. doi:10.1109/TBME.2020.3043215. URL <https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=9286420>
- [32] M. M. Hasan, J. Phu, H. Wang, A. Sowmya, M. Kalloniatis, E. Meijering, H. MM, P. J, W. H, S. A, K. M, M. E, Oct-based diagnosis of glaucoma and glaucoma stages using explainable machine learning, *Scientific reports* 15 (1) (2025) 3592. doi:10.1038/s41598-025-87219-w. URL <https://pubmed.ncbi.nlm.nih.gov/39875492/>
- [33] G. Aresta, T. Araújo, U. Schmidt-Erfurth, H. Bogunovic, A. G, A. T, S.-E, U, B, H, Anomaly detection in retinal oct images with deep learning-based knowledge distillation, *Translational vision science & technology* 14 (3) (2025) 26. doi:10.1167/tvst.14.3.26. URL <https://pubmed.ncbi.nlm.nih.gov/40146150/>
- [34] S. Gurumoorthy, N. Meena Lochana Nayaki, K. E. Vendhan, G. S., V. K.E., The role of artificial intelligence in monitoring glaucoma progression using optical coherence tomography, *Indian Journal of Clinical and Experimental Ophthalmology* 11 (4) (2025) 631–640. doi:10.18231/j.ijceo.11108.1761806059. URL <https://www.scopus.com/pages/publications/105026345646?origin=resultslist>
- [35] M. Al-Zubaidy, A. Stankiewicz, M. Anderson, J. Reed, V. Corona, R. Pope, B. Obara, M. S. Habib, D. H. Steel, A.-Z. M., S. A., A. M., R. J., C. V., P. R., O. B., H. M.S., S. D.H., A scoping review of the use of artificial intelligence models in automated oct analysis and prediction of treatment outcomes in diabetic macular oedema, *Informatics in Medicine Unlocked* 57 (2025). doi:10.1016/j.imu.2025.101676. URL <https://www.scopus.com/pages/publications/105011972572?origin=resultslist>
- [36] X. Wei, R. Sui, W. X, S. R, A review of machine learning algorithms for retinal cyst segmentation on optical coherence tomography, *Sensors (Basel, Switzerland)* 23 (6) (2023). doi:10.3390/s23063144. URL <https://pubmed.ncbi.nlm.nih.gov/36991857/>
- [37] A. Hayati, M. R. Abdol Homayuni, R. Sadeghi, H. Asadigandomani, M. Dashtkoobi, S. Eslami, M. Soleimani, H. A, A. H. MR, S. R, A. H, D. M, E. S, S. M, A. H. M.R., Advancing diabetic retinopathy screening: A systematic review of artificial intelligence and optical coherence tomography angiography innovations, *Diagnostics (Basel, Switzerland)* 15 (6) (2025). doi:10.3390/diagnostics15060737. URL <https://pubmed.ncbi.nlm.nih.gov/40150080/>
- [38] D. B. Russakoff, S. S. Mannil, J. D. Oakley, A. R. Ran, C. Y. Cheung, S. Dasari, M. Riyazzuddin, S. Nagaraj, H. L. Rao, D. Chang, R. T. Chang, R. DB, M. SS, O. JD, R. AR, C. CY, D. S, R. M, N. S, R. HL, C. D, C. RT, R. D.B., M. S.S., O. J.D., R. A.R., C. C.Y., R. H.L., C. R.T., A 3d deep learning system for detecting referable glaucoma using full oct macular

- cube scans, *Translational vision science & technology* 9 (2) (2020) 12. doi:10.1167/tvst.9.2.12. URL <https://pubmed.ncbi.nlm.nih.gov/32704418/>
- [39] L.-C. Sun, S.-I. Pao, K.-H. Huang, C.-Y. Wei, K.-F. Lin, P.-N. Chen, S. L.C., P. SI, H. KH, W. CY, L. KF, C. PN, S. L.-C., P. S.-I., H. K.-H., W. C.-Y., L. K.-F., C. P.-N., Generative adversarial network-based deep learning approach in classification of retinal conditions with optical coherence tomography images, *Graefes Archive for Clinical and Experimental Ophthalmology* 261 (5) (2023) 1399–1412. doi:10.1007/s00417-022-05919-9. URL <https://pubmed.ncbi.nlm.nih.gov/36441228/>
- [40] J. Lo, T. T. Yu, D. Ma, P. Zang, J. P. Owen, Q. Zhang, R. K. Wang, M. F. Beg, A. Y. Lee, Y. Jia, M. V. Sarunic, Federated learning for microvasculature segmentation and diabetic retinopathy classification of OCT data, *Ophthalmology Science* 1 (4) (2021) 100069. doi:10.1016/j.xops.2021.100069. URL <https://doi.org/10.1016/j.xops.2021.100069>
- [41] D. Rivas-Villar, A. R. Motschi, M. Pircher, C. K. Hitzenberger, M. Schranz, P. K. Roberts, U. Schmidt-Erfurth, H. Bogunović, R.-V. D. M. AR, P. M. H. CK, S. M. R. PK, S.-E. U, B. H. M. A.R., H. C.K., R. P.K., Automated inter-device 3d oct image registration using deep learning and retinal layer segmentation, *Biomedical optics express* 14 (7) (2023) 3726–3747. doi:10.1364/BOE.493047. URL <https://pubmed.ncbi.nlm.nih.gov/37497506/>
- [42] T. Schlosser, F. Beuth, T. Meyer, A. S. Kumar, G. Stolze, O. Furashova, K. Engelmann, D. Kowanko, Visual Acuity Prediction on Real-Life Patient Data Using a Machine Learning Based Multistage System, *Scientific Reports* 14 (1) (2024) 5532. doi:10.1038/s41598-024-54482-2. URL <https://www.nature.com/articles/s41598-024-54482-2>
- [43] Health Level Seven International, FHIR release 4 (version 4.0.1), HL7 International (2019). URL <https://hl7.org/fhir/R4/>
- [44] DICOM Standards Committee, DICOM PS3.1 2025e: Introduction and overview, National Electrical Manufacturers Association (2025). URL <https://dicom.nema.org/medical/dicom/2025e/output/html/part01.html>
- [45] K. Tavakoli, F. G. P. Kalaw, S. Bhanvadia, M. Hogarth, S. L. Baxter, Concept coverage analysis of ophthalmic infections and trauma among the standardized medical terminologies SNOMED-CT, ICD-10-CM, and ICD-11, *Ophthalmology Science* 3 (4) (2023) 100337. doi:10.1016/j.xops.2023.100337. URL <https://doi.org/10.1016/j.xops.2023.100337>
- [46] F. G. P. Kalaw, J. S. Chen, S. L. Baxter, Variations in using diagnosis codes for defining age-related macular degeneration cohorts, *Informatics* 11 (2) (2024) 28. doi:10.3390/informatics11020028. URL <https://doi.org/10.3390/informatics11020028>
- [47] A. S. Tejani, M. E. Klontzas, A. A. Gatti, J. T. Mongan, L. Moy, S. H. Park, C. E. Kahn, Jr., CLAIM 2024 Update Panel, Checklist for artificial intelligence in medical imaging (CLAIM): 2024 update, *Radiology: Artificial Intelligence* 6 (4) (2024) e240300. doi:10.1148/ryai.240300. URL <https://doi.org/10.1148/ryai.240300>
- [48] A. Aytulun, A. Cruz-Herranz, O. Aktas, L. J. Balcer, L. Balk, P. Barboni, A. A. Blanco, P. A. Calabresi, F. Costello, B. Sanchez-Dalmau, D. C. DeBuc, N. Feltgen, R. P. Finger, J. L. Frederiksen, E. Frohman, T. Frohman, D. Garway-Heath, I. Gabilondo, J. S. Graves, A. J. Green, H. P. Hartung, J. Havla, F. G. Holz, J. Imitola, R. Kenney, A. Klistorner, B. Knier, T. Korn, S. Kolbe, J. Krämer, W. A. Lagrèze, L. Leocani, O. Maier, E. H. Martínez-Lapiscina, S. Meuth, O. Outteryck, F. Paul, A. Petzold, G. Pihl-Jensen, J. L. Preiningerova, G. Rebolledo, M. Ringelstein, S. Saidha, S. Schippling, J. S. Schuman, R. C. Sergott, A. Toosy, P. Villoslada, S. Wolf, E. A. Yeh, P. Yu-Wai-Man, H. G. Zimmermann, A. U. Brandt, P. Albrecht, APOSTEL 2.0 recommendations for reporting quantitative optical coherence tomography studies, *Neurology* 97 (2) (2021) 68–79. doi:10.1212/WNL.00000000000012125. URL <https://doi.org/10.1212/WNL.00000000000012125>
- [49] J. E. Alderman, J. Palmer, E. Laws, M. D. McCradden, J. Ordish, M. Ghassemi, S. R. Pfohl, N. Rostamzadeh, H. Cole-Lewis, B. Glocker, M. Calvert, T. J. Pollard, J. Gill, J. Gath, A. Adebajo, J. Beng, C. H. Leung, S. Kuku, L.-A. Farmer, R. N. Matin, B. A. Mateen, F. McKay, K. Heller, A. Karthikesalingam, D. Treanor, M. Mackintosh, L. Oakden-Rayner, R. Pearson, A. K. Manrai, P. Myles, J. Kumuthini, Z. Kapacee, N. J. Sebire, L. H. Nazer, J. Seah, A. Akbari, L. Berman, J. W. Gichoya, L. Righetto, D. Samuel, W. Wasswa, M. Charalambides, A. Arora, S. Pujari, C. Summers, E. Sapey, S. Wilkinson, V. Thakker, A. Denniston, X. Liu, Tackling algorithmic bias and promoting transparency in health datasets: the STANDING Together consensus recommendations, *The Lancet Digital Health* 7 (1) (2025) e64–e88. doi:10.1016/S2589-7500(24)00224-3. URL [https://doi.org/10.1016/S2589-7500\(24\)00224-3](https://doi.org/10.1016/S2589-7500(24)00224-3)
- [50] R. K. Parrish, II, L. R. Pasquale, A. Y. Lee, R. Folberg, M. W. Stewart, S. L. Duncan Powers, Who could know who I am? the possibility of patient identification with retinal imaging, *American Journal of Ophthalmology* 216 (2020) A3–A4. doi:10.1016/j.ajo.2020.03.039. URL <https://doi.org/10.1016/j.ajo.2020.03.039>
- [51] G. Nebbia, S. Kumar, S. M. McNamara, C. Bridge, J. P. Campbell, M. F. Chiang, N. Mandava, P. Singh, J. Kalpathy-Cramer, Re-identification of patients from imaging features extracted by foundation models, *npj Digital Medicine* 8 (1) (2025) 469. doi:10.1038/s41746-025-01801-0. URL <https://doi.org/10.1038/s41746-025-01801-0>
- [52] Z. L. Teo, X. Zhang, Y. Yang, L. Jin, C. Zhang, S. S. J. Poh, W. Yu, Y. Chen, J. B. Jonas, Y. X. Wang, W. C. Wu, C. C. Lai, Y. Liu, R. S. M. Goh, D. S. W. Ting, Privacy-preserving technology using federated learning and blockchain in protecting against adversarial attacks for retinal imaging, *Ophthalmology* 132 (4) (2025) 484–494. doi:10.1016/j.opht.2024.10.017. URL <https://doi.org/10.1016/j.opht.2024.10.017>
- [53] T. Schlegl, S. M. Waldstein, H. Bogunovic, F. Endstraßer, A. Sadeghipour, A.-M. Philip, D. Podkowinski, B. S. Gerendas, G. Langs, U. Schmidt-Erfurth, Fully automated detection and quantification of macular fluid in OCT using deep learning, *Ophthalmology* 125 (4) (2018) 549–558. doi:10.1016/j.opht.2017.10.031. URL <https://doi.org/10.1016/j.opht.2017.10.031>
- [54] A. Szeskin, R. Yehuda, O. Shmueli, J. Levy, L. Joskowicz, S. A. Y. R. S. O. L. J. J. L., A column-based deep learning method for the detection and quantification of atrophy associated with amd in oct scans, *Medical image analysis* 72 (2021) 102130. doi:10.1016/j.media.2021.102130. URL <https://pubmed.ncbi.nlm.nih.gov/34198041/>
- [55] T. D. L. Keenan, T. E. Clemons, A. Domalpally, M. J. Elman, M. Havilio, E. Agrón, G. Benyamini, E. Y. Chew, K. TDL, C. TE, D. A. E. MJ, H. M. A. E, B. G. C. EY, K. T.D.L., C. T.E., E. M.J., C. E.Y., Retinal specialist versus artificial intelligence detection of retinal fluid from oct: Age-related eye disease study 2: 10-year follow-on study, *Ophthalmology* 128 (1) (2021) 100–109. doi:10.1016/j.opht.2020.06.038. URL <https://pubmed.ncbi.nlm.nih.gov/32598950/>
- [56] M. D. Abramoff, P. T. Lavin, M. Birch, N. Shah, J. C. Folk, Pivotal trial of an autonomous AI-based diagnostic system for detection of diabetic retinopathy in primary care offices, *npj Digital Medicine* 1 (2018) 39. doi:10.1038/s41746-018-0040-6. URL <https://doi.org/10.1038/s41746-018-0040-6>
- [57] P. Ruamviboonsuk, R. Tiwari, R. Sayres, V. Nganthavee, K. Hemarat, A. Kongprayoon, R. Raman, B. Levinstein, Y. Liu, M. Schaeckermann, R. Lee, S. Virmani, K. Widner, J. Chambers, F. Hersch, L. Peng, D. R. Webster, Real-time diabetic retinopathy screening by deep learning in a multisite national screening programme: a prospective interventional cohort study, *The Lancet Digital Health* 4 (4) (2022) e235–e244. doi:10.1016/S2589-7500(22)00017-6. URL [https://doi.org/10.1016/S2589-7500\(22\)00017-6](https://doi.org/10.1016/S2589-7500(22)00017-6)
- [58] Y. Xie, Q. D. Nguyen, H. Hamzah, G. Lim, V. Bellemo, D. V. Gunasekaran, M. Y. T. Yip, X. Q. Lee, W. Hsu, M. L. Lee, C. S. Tan, H. T. Wong, E. L.

- Lamoureux, G. S. W. Tan, T. Y. Wong, E. A. Finkelstein, D. S. W. Ting, Artificial intelligence for teleophthalmology-based diabetic retinopathy screening in a national programme: an economic analysis modelling study, *The Lancet Digital Health* 2 (5) (2020) e240–e249. doi:10.1016/S2589-7500(20)30060-1. URL [https://doi.org/10.1016/S2589-7500\(20\)30060-1](https://doi.org/10.1016/S2589-7500(20)30060-1)
- [59] B. Chuter, V. Joshi, S. Hallaj, E. Walker, C. Bowd, A. Belghith, M. H. Goldbaum, A. Grzybowski, M. A. Fazio, C. A. Girkin, C. G. De Moraes, J. M. Liebmann, R. N. Weinreb, L. M. Zangwill, M. Christopher, C. B. J. V. H. S. W. E. B. C. B. A. G. MH, G. A. F. MA, G. CA, D. M. CG, L. JM, W. RN, Z. LM, C. M, G. M.H., F. M.A., G. C.A., D. M. C.G., L. J.M., W. R.N., Z. L.M., A novel multimodal implementation of a foundation artificial intelligence model using optic nerve head fundus photographs and oct imaging for glaucoma detection, *Ophthalmology science* 6 (2) (2026) 101012. doi:10.1016/j.xops.2025.101012. URL <https://pubmed.ncbi.nlm.nih.gov/41631246/>
- [60] R. Liu, Q. Li, F. Xu, S. Wang, J. He, Y. Cao, F. Shi, X. Chen, J. Chen, L. R. L. Q. X. F. W. S. H. J. C. Y. S. F. C. X. C. J, Application of artificial intelligence-based dual-modality analysis combining fundus photography and optical coherence tomography in diabetic retinopathy screening in a community hospital, *Biomedical engineering online* 21 (1) (2022) 47. doi:10.1186/s12938-022-01018-2. URL <https://pubmed.ncbi.nlm.nih.gov/35859144/>
- [61] M. D. Abramoff, N. Whitestone, J. L. Patnaik, E. Rich, M. Ahmed, L. Husain, M. Y. Hassan, M. S. H. Tanjil, D. Weitzman, T. Dai, B. D. Wagner, D. H. Cherwek, N. Congdon, K. Islam, Autonomous artificial intelligence increases real-world specialist clinic productivity in a cluster-randomized trial, *npj Digital Medicine* 6 (1) (2023) 184. doi:10.1038/s41746-023-00931-7. URL <https://doi.org/10.1038/s41746-023-00931-7>
- [62] K. Hou, C. Bradley, P. Herbert, C. Johnson, M. Wall, P. Y. Ramulu, M. Unberath, J. Yohannan, H. K. B. C. H. P. J. C. W. M. R. PY, U. M. Y. J, Predicting visual field worsening with longitudinal oct data using a gated transformer network, *Ophthalmology* 130 (8) (2023) 854–862. doi:10.1016/j.ophtha.2023.03.019. URL <https://pubmed.ncbi.nlm.nih.gov/37003520/>
- [63] J. Jung, J. Han, J. M. Han, J. Ko, J. Yoon, J. S. Hwang, J. I. Park, G. Hwang, J. H. Jung, D. D.-J. Hwang, J. J. H. J. H. JM, K. J. Y. J. H. JS, P. JI, H. G, J. JH, H. DD, H. J.M., H. J.S., P. J.I., J. J.H., H. D.D.-J., Prediction of neovascular age-related macular degeneration recurrence using optical coherence tomography images with a deep neural network, *Scientific reports* 14 (1) (2024) 5854. doi:10.1038/s41598-024-56309-6. URL <https://pubmed.ncbi.nlm.nih.gov/38462646/>
- [64] L. Wei, W. He, J. Wang, K. Zhang, Y. Du, J. Qi, J. Meng, X. Qiu, L. Cai, Q. Fan, Z. Zhao, Y. Tang, S. Ni, H. Guo, Y. Song, X. He, D. Ding, Y. Lu, X. Zhu, W. L. H. W. W. J. Z. K. D. Y. Q. J. M. J. Q. X. C. L. F. Q. Z. Z. T. Y, N. S, G. H, S. Y, H. X, D. D, L. Y, Z. X, An optical coherence tomography-based deep learning algorithm for visual acuity prediction of highly myopic eyes after cataract surgery, *Frontiers in cell and developmental biology* 9 (2021) 652848. doi:10.3389/fcell.2021.652848. URL <https://pubmed.ncbi.nlm.nih.gov/34124042/>
- [65] D. A. Edwards, B. Emerick, A. G. Kondic, K. Kiradjiev, C. Raymond, M. Zyskin, Mathematical models for the effect of anti-vascular endothelial growth factor on visual acuity, *Journal of Mathematical Biology* 81 (6) (2020) 1397–1428. doi:10.1007/s00285-020-01544-4. URL <https://link.springer.com/article/10.1007/s00285-020-01544-4>
- [66] G. S. Collins, K. G. M. Moons, P. Dhiman, R. D. Riley, A. L. Beam, B. Van Calster, M. Ghassemi, X. Liu, J. B. Reitsma, M. van Smeden, A.-L. Boulesteix, J. C. Camaradou, L. A. Celi, S. Denaxas, A. K. Denniston, B. Glocker, R. M. Golub, H. Harvey, G. Heinze, M. M. Hoffman, A. P. Kengne, E. Lam, N. Lee, E. W. Loder, L. Maier-Hein, B. A. Mateen, M. D. McCradden, L. Oakden-Rayner, J. Ordish, R. Parnell, S. Rose, K. Singh, L. Wynants, P. Logullo, TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods, *BMJ* 385 (2024) e078378. doi:10.1136/bmj-2023-078378. URL <https://doi.org/10.1136/bmj-2023-078378>
- [67] A. Feinberg, E. Berger, R. Hammond, 2023 healthcare provider IT report: Doubling down on innovation, Bain & Company (Sep. 2023). URL <https://www.bain.com/insights/2023-healthcare-provider-it-report-doubling-down-on-innovation/>
- [68] E. Berger, C. Dowling, A. Feinberg, R. Hammond, Healthcare IT spending: Innovation, integration, and AI, Bain & Company (Sep. 2024). URL <https://www.bain.com/insights/healthcare-it-spending-innovation-integration-ai/>
- [69] A. Tyson, G. Pasquini, A. Spencer, C. Funk, 60% of Americans would be uncomfortable with provider relying on AI in their own health care, Pew Research Center (Feb. 2023). URL <https://www.pewresearch.org/science/2023/02/22/60-of-americans-would-be-uncomfortable-with-provider-relying-on-ai-in-their-own-health-care/>
- [70] X. Chen, J. Wang, T. Qian, J. Wang, Y. Ding, S. Zhang, J. Liao, Q. Cheng, T. Yang, M. Mateen, Y. Fan, Z. Song, J. Chen, S. Li, J. Hu, W. Yan, H. Chen, W. Wu, J. Huang, T. Y. Wong, X. Xu, C. X. W. J. Q. T. D. Y. Z. S. L. J. C. Q. Y. T. M. M. F. Y. S. Z. C. J. L. S. H. J. Y. W. C. H. W. W. H. J. W. TY, X. X, W. T.Y., An artificial intelligence cloud platform for oct-based retinal anomalies screening system in real clinical environments, *NPJ digital medicine* 8 (1) (2025) 559. doi:10.1038/s41746-025-01959-7. URL <https://pubmed.ncbi.nlm.nih.gov/40883412/>
- [71] E. Beede, E. Baylor, F. Hersch, A. Iurchenko, L. Wilcox, P. Ruamviboonsuk, L. M. Vardoulakis, A human-centered evaluation of a deep learning system deployed in clinics for the detection of diabetic retinopathy, in: *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*, Association for Computing Machinery, 2020, pp. 1–12. doi:10.1145/3313831.3376718. URL <https://doi.org/10.1145/3313831.3376718>
- [72] M. Ghassemi, L. Oakden-Rayner, A. L. Beam, The false hope of current approaches to explainable artificial intelligence in health care, *The Lancet Digital Health* 3 (11) (2021) e745–e750. doi:10.1016/S2589-7500(21)00208-9. URL [https://doi.org/10.1016/S2589-7500\(21\)00208-9](https://doi.org/10.1016/S2589-7500(21)00208-9)
- [73] N. Arun, N. Gaw, P. Singh, K. Chang, M. Aggarwal, B. Chen, K. Hoebel, S. Gupta, J. Patel, M. Gidwani, J. Adebayo, M. D. Li, J. Kalpathy-Cramer, Assessing the trustworthiness of saliency maps for localizing abnormalities in medical imaging, *Radiology: Artificial Intelligence* 3 (6) (2021) e200267. doi:10.1148/ryai.2021200267. URL <https://doi.org/10.1148/ryai.2021200267>
- [74] S. Gaube, H. Suresh, M. Raue, A. Merritt, S. J. Berkowitz, E. Lerner, J. F. Coughlin, J. V. Guttag, E. Colak, M. Ghassemi, Do as AI say: susceptibility in deployment of clinical decision-aids, *npj Digital Medicine* 4 (1) (2021) 31. doi:10.1038/s41746-021-00385-9. URL <https://doi.org/10.1038/s41746-021-00385-9>
- [75] European Parliament and Council of the European Union, Regulation (EU) 2024/1689 of the European Parliament and of the Council of 13 June 2024 laying down harmonised rules on artificial intelligence and amending regulations (EC) no 300/2008, (EU) no 167/2013, (EU) no 168/2013, (EU) 2018/858, (EU) 2018/1139 and (EU) 2019/2144 and directives 2014/90/EU, (EU) 2016/797 and (EU) 2020/1828 (artificial intelligence act) (text with EEA relevance), *Official Journal of the European Union*, L 2024/1689, 12 July 2024 (2024). URL <https://data.europa.eu/eli/reg/2024/1689/oj>